

Optimal Execution

BayQuant

The Trading Dilemma

Liquidating a large position is no free lunch: execution moves prices against the order. Sell it all at once and **market impact** is concentrated into a short, severe shock. Spread it out and impact cost falls, but the remaining shares sit exposed to the market's random swings for longer: **timing risk**. This report replicates Almgren and Chriss [1], who cast optimal liquidation as minimizing expected cost plus a constant times its variance.

Price Dynamics and Impact

The price follows Arithmetic Brownian Motion, disturbed by volatility, drift, and the trader's own activity. Discretizing the horizon into N steps of length $\tau = T/N$, with x_k shares held after step k ($x_0 = X$, $x_N = 0$) and $n_k = x_{k-1} - x_k$ shares traded in step k , the price process is

$$S_k = S_{k-1} + \alpha\tau + \sigma\sqrt{\tau}\xi_k - \tau g\left(\frac{n_k}{\tau}\right), \quad (1)$$

where ξ_k are i.i.d. shocks with zero mean and unit variance, $g(v)$ is the **permanent impact** (persists in the price process), and the drift α is taken as **zero** throughout since no directional information is assumed. Trading also incurs a **temporary impact** $h(v)$: a transient concession on the execution price of a single step, so the price actually received on sale k is

$$\tilde{S}_k = S_{k-1} - h\left(\frac{n_k}{\tau}\right).$$

In both g and h , the argument $v = n_k/\tau$ is the average rate of trading during the interval t_{k-1} to t_k .

Cost of Trading

Following Perold (1988) [2], the **implementation shortfall** $XS_0 - \sum_k n_k \tilde{S}_k$ (2) is the cost of trading relative to the initial book value. Expanding the capture term $\sum_k n_k \tilde{S}_k$ (total trading revenue) gives the shortfall's expectation and variance

$$E(x) = \sum_k \tau x_k g\left(\frac{n_k}{\tau}\right) + \sum_k n_k h\left(\frac{n_k}{\tau}\right),$$

$$V(x) = \sigma^2 \sum_k \tau x_k^2.$$

Specializing to linear impact, $g(v) = \gamma v$ and $h(v) = \varepsilon \operatorname{sgn}(v) + \eta v$ (with ε a fixed cost per trade, e.g. half the bid-ask spread), summation by parts gives the permanent-impact term

$$\sum_k \tau x_k g\left(\frac{n_k}{\tau}\right) = \frac{1}{2}\gamma X^2 - \frac{1}{2}\gamma \sum_k n_k^2, \quad (3)$$

so its contribution to $E(x)$ is **not** schedule-independent at finite τ . Temporary impact contributes

$$\sum_k n_k h\left(\frac{n_k}{\tau}\right) = \varepsilon \sum_k |n_k| + (\eta/\tau) \sum_k n_k^2.$$

Combining, and writing $\bar{\eta} = \eta - \frac{1}{2}\gamma\tau$ to absorb the schedule-dependent permanent-impact term,

$$E(x) = \frac{1}{2}\gamma X^2 + \varepsilon \sum_k |n_k| + \frac{\bar{\eta}}{\tau} \sum_k n_k^2.$$

For a monotone schedule (n_k all one sign), $\sum_k |n_k| = X$, so only the last term shapes the schedule. Almgren and Chriss minimize the utility function

$$U(x) = E(x) + \lambda V(x)$$

over $x_1, \dots, x_{N-1}$, where $\lambda \geq 0$ is risk aversion.

The Optimal Schedule

Dropping the schedule-independent terms $\frac{1}{2}\gamma X^2 + \varepsilon X$ and writing $a = \bar{\eta}/\tau$, $b = \lambda\sigma^2\tau$ turns the objective into the quadratic form

$$U = a \sum_k (x_{k-1} - x_k)^2 + b \sum_k x_k^2,$$

with the endpoints $x_0 = X_0$ and $x_N = X_T$ held fixed and only the interior holdings $x_1, \dots, x_{N-1}$ free.

Stationarity condition (the gradient vanishes, $\nabla U = 0$). U is a sum of squares, a convex bowl with a single minimum where every partial derivative vanishes. For an interior x_j , the only terms containing it are $(x_{j-1} - x_j)^2$, $(x_j - x_{j+1})^2$, and $b x_j^2$, so

$$\frac{\partial U}{\partial x_j} = a[-2(x_{j-1} - x_j) + 2(x_j - x_{j+1})] + 2bx_j = 0.$$

Dividing by 2 and rearranging gives:

$$x_{j+1} - 2x_j + x_{j-1} = \frac{b}{a}x_j = \frac{\lambda\sigma^2}{\bar{\eta}}\tau^2 x_j = \tilde{\kappa}^2\tau^2 x_j.$$

Linear difference equation. Dividing by τ^2 exposes a centered second difference,

$$\frac{x_{j+1} - 2x_j + x_{j-1}}{\tau^2} = \tilde{\kappa}^2 x_j, \quad t_j = j\tau.$$

This is a linear, constant-coefficient, second-order difference equation with values fixed at both ends, $x_0 = X_0$ and $x_N = X_T$. Liquidation sets $X_0 = X$ and $X_T = 0$, but general endpoints cost nothing and also cover re-planning partway through a trade. A second-order equation has a two-dimensional solution space, so the task is to find two independent solutions and fit two constants.

Characteristic equation and roots. Substituting the trial solution $x_j = e^{\theta t_j}$, the shifted terms are $x_{j\pm 1} = e^{\theta t_j} e^{\pm\theta\tau}$, so the left-hand side becomes

$$\frac{1}{\tau^2} e^{\theta t_j} (e^{-\theta\tau} - 2 + e^{\theta\tau}) = \frac{2}{\tau^2} (\cosh(\theta\tau) - 1) e^{\theta t_j},$$

using $e^{\theta\tau} + e^{-\theta\tau} = 2 \cosh(\theta\tau)$. The factor $e^{\theta t_j}$ is never zero, so it cancels, and the recurrence holds for every j exactly when

$$\frac{2}{\tau^2} (\cosh(\theta\tau) - 1) = \tilde{\kappa}^2.$$

This **characteristic equation** fixes the rate θ from the input $\tilde{\kappa}$: the known $\tilde{\kappa}$ on the right, the unknown θ inside the cosh on the left. The two differ because a second difference is not a second derivative: $2(\cosh z - 1) = z^2 + z^4/12 + \dots$, so $\theta = \tilde{\kappa} + O(\tau^2)$. Because cosh is even, $-\theta$ solves it whenever θ does, so the recurrence has the two solutions $e^{\theta t_j}$ and $e^{-\theta t_j}$; their ratio $e^{2\theta t_j}$ is non-constant for $\theta > 0$, so they are linearly independent, and since a second-order equation has a two-dimensional solution space they span every solution.

General solution and boundary conditions. Every solution is therefore a combination of the two, with two free constants,

$$x_j = Ae^{\theta t_j} + Be^{-\theta t_j}.$$

The endpoints pin A and B : at $t_0 = 0$, $x_0 = A + B = X_0$, and at $t_N = T$, $x_N = Ae^{\theta T} + Be^{-\theta T} = X_T$. Solving this pair and reassembling collapses the exponentials into hyperbolic sines (the algebra is in the Appendix),

$$x_j = \frac{X_0 \sinh(\theta(T - t_j)) + X_T \sinh(\theta t_j)}{\sinh(\theta T)},$$

a hyperbolic interpolation between the two endpoints: the first term carries the start value X_0 and vanishes at $t = T$, the second carries X_T and vanishes at $t = 0$, and $\sinh(\theta T)$ normalizes both. As $\theta \rightarrow 0$ it reduces to the straight line $x_j = (X_0(T - t_j) + X_T t_j)/T$, and larger θ bows the path further from it. Setting $X_0 = X$ and $X_T = 0$ gives the liquidation trajectory

$$x_j = X \frac{\sinh(\theta(T - t_j))}{\sinh(\theta T)}, \quad (4)$$

for $j = 0, \dots, N$, and the associated trade list

$$n_j = \frac{2 \sinh(\frac{1}{2}\theta\tau)}{\sinh(\theta T)} \cosh(\theta(T - t_{j-\frac{1}{2}})) X, \quad (5)$$

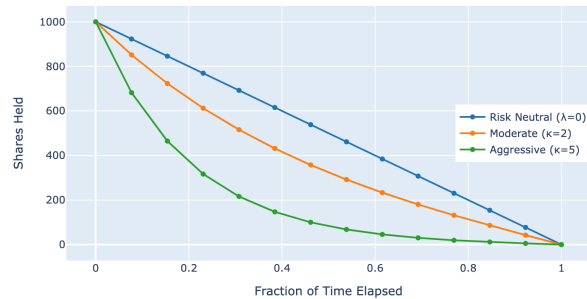
for $j = 1, \dots, N$, where $t_{j-\frac{1}{2}} = (j - \frac{1}{2})\tau$ is the midpoint of step j .

Continuous-time limit. As $\tau \rightarrow 0$, $\bar{\eta} \rightarrow \eta$ and both θ and $\tilde{\kappa}$ tend to $\kappa = \sqrt{\lambda\sigma^2/\eta}$, so (4) becomes $x(t) = X \sinh(\kappa(T - t))/\sinh(\kappa T)$, the solution of $\ddot{x} = \kappa^2 x$, and the trading rate n_j/τ in (5) becomes $-\dot{x}(t)$.

The Shape of the Schedule

θ alone governs the curve's shape. As $\lambda \rightarrow 0$, $\theta \rightarrow 0$ and $\sinh(z) \approx z$ makes the schedule linear: constant-rate, TWAP-like execution. As λ grows, θ grows and the curve front-loads, trading more impact cost for less timing risk. $\frac{1}{\theta}$ is the characteristic unwind timescale.

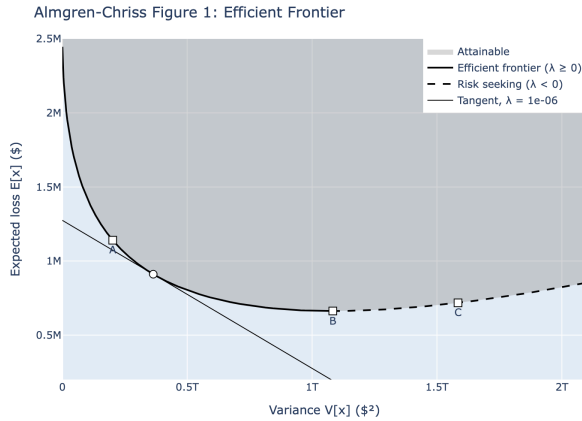
Optimal Holdings Trajectory x_j



The Efficient Frontier

Sweeping $\lambda \geq 0$ traces out a curve of $(V(x), E(x))$ pairs, one per optimal trajectory: the **efficient frontier** of optimal execution. Each point is the lowest expected cost achievable at that variance; no monotone schedule beats the frontier on both at once. At $\lambda = 0$ the frontier sits at its minimum-cost, maximum-variance end (the TWAP schedule, which minimizes $\sum_k n_k^2$ for fixed X); increasing λ moves along the frontier toward lower variance at higher cost as the schedule front-loads.

The figure replicates Figure 1 of [1] with their test-case parameters ($X = 10^6$ shares, $T = 5$ days, $N = 5$). Past the risk-neutral point B, $\lambda < 0$ (dashed) describes a risk-seeking trader who delays selling and pays more in both cost and variance, so that branch is not efficient. The straight line is tangent at $\lambda = 10^{-6}$ with slope $-\lambda$.



Half-Life of a Trade

For $\theta T \gtrsim 1$, $\sinh(\theta(T-t))/\sinh(\theta T) \approx e^{-\theta t}$, so holdings decay approximately exponentially, $x(t) \approx X e^{-\theta t}$. The time for the position to fall to half its initial size is the **half-life**

$$t_{1/2} = \frac{\ln(2)}{\theta} \approx \frac{\ln(2)}{\kappa} = \ln(2) \sqrt{\frac{\eta}{\lambda \sigma^2}}.$$

Half-life shrinks with risk aversion λ and volatility σ (faster unwinds when risk is costlier) and grows with the temporary-impact coefficient η (slower unwinds when trading is more expensive).

Buying as the Mirror Image

Since cost enters only through n_k^2 , the objective is direction-blind: accumulating a position just flips the boundary conditions to $x_0 = 0$, $x_N = X$, giving the time-reversal of the sell schedule,

$$x_j = X \frac{\sinh(\theta t_j)}{\sinh(\theta T)}.$$

References

- [1] R. Almgren and N. Chriss, "Optimal execution of portfolio transactions," *Journal of Risk*, vol. 3, no. 2, pp. 5–39, 2000.
- [2] A. F. Perold, "The implementation shortfall: Paper versus reality," *Journal of Portfolio Management*, vol. 14, no. 3, pp. 4–9, 1988.

Appendix

Price Process from Continuous Time

Equation (1)

$$dS_t = (\alpha - g(v_t)) dt + \sigma dW_t,$$

where v_t is the trader's continuous trading rate. Discretizing via Euler-Maruyama, with n_k/τ standing in for v_t and $W_{t_k} - W_{t_{k-1}} \approx \sqrt{\tau}\xi_k$ for the Brownian increment over step k , gives

$$S_k = S_{k-1} + \alpha\tau + \sigma\sqrt{\tau}\xi_k - \tau g\left(\frac{n_k}{\tau}\right), \quad k = 1, \dots, N.$$

Capture Identity

Equation (2)

Substituting the price dynamics and temporary-impact price into the definition of capture (total trading revenue), $\sum_k n_k \tilde{S}_k$, and expanding gives equation (2):

$$\sum_{k=0}^N n_k \tilde{S}_k = XS_0 + \sum_{k=1}^N \left(\sigma\sqrt{\tau}\xi_k - \tau g\left(\frac{n_k}{\tau}\right) \right) x_k - \sum_{k=1}^N n_k h\left(\frac{n_k}{\tau}\right).$$

Permanent-Impact Summation by Parts

Equation (3)

Since $n_k = x_{k-1} - x_k$,

$$x_{k-1}^2 - x_k^2 = (x_{k-1} - x_k)(x_{k-1} + x_k) = n_k x_{k-1} + n_k x_k.$$

Summing over $k = 1, \dots, N$ telescopes the left side to $x_0^2 - x_N^2 = X^2$, giving

$$\sum_k n_k x_{k-1} + \sum_k n_k x_k = X^2.$$

Separately, $n_{k(x_{k-1}-x_k)} = n_k^2$, so

$$\sum_k n_k x_{k-1} - \sum_k n_k x_k = \sum_k n_k^2.$$

Subtracting the second identity from the first and dividing by 2,

$$\sum_k n_k x_k = \frac{1}{2}X^2 - \frac{1}{2}\sum_k n_k^2.$$

Multiplying by γ gives equation (3).

Solving the Optimal-Schedule Recurrence

Equation (4)

Substituting $x_j = e^{\theta t_j}$ into the recurrence, with $t_{j\pm 1} = t_j \pm \tau$,

$$\frac{e^{\theta\tau} - 2 + e^{-\theta\tau}}{\tau^2} e^{\theta t_j} = \frac{2}{\tau^2} (\cosh(\theta\tau) - 1) e^{\theta t_j} = \tilde{\kappa}^2 e^{\theta t_j},$$

so $e^{\theta t_j}$ is an exact solution whenever θ satisfies the cosh relation, and by evenness so is $e^{-\theta t_j}$. The general solution is their combination

$$x_j = Ae^{\theta t_j} + Be^{-\theta t_j},$$

and the two endpoints give a linear system in A and B ,

$$A + B = X_0, \quad Ae^{\theta T} + Be^{-\theta T} = X_T.$$

From the first equation $B = X_0 - A$; substituting into the second and using $e^{\theta T} - e^{-\theta T} = 2 \sinh(\theta T)$,

$$A = \frac{X_T - X_0 e^{-\theta T}}{2 \sinh(\theta T)}, \quad B = \frac{X_0 e^{\theta T} - X_T}{2 \sinh(\theta T)}.$$

Reassembling $x_j = Ae^{\theta t_j} + Be^{-\theta t_j}$ and grouping the two coefficients,

$$\begin{aligned} 2 \sinh(\theta T) x_j &= X_T (e^{\theta t_j} - e^{-\theta t_j}) + X_0 (e^{\theta T} e^{-\theta t_j} - e^{-\theta T} e^{\theta t_j}) \\ &= X_T (e^{\theta t_j} - e^{-\theta t_j}) + X_0 (e^{\theta(T-t_j)} - e^{-\theta(T-t_j)}) \\ &= 2X_T \sinh(\theta t_j) + 2X_0 \sinh(\theta(T-t_j)), \end{aligned}$$

where the second line used $e^{\theta T} e^{-\theta t_j} = e^{\theta(T-t_j)}$ and $e^{-\theta T} e^{\theta t_j} = e^{-\theta(T-t_j)}$. Each $\sinh$ emerges on its own from a pair of exponentials of opposite sign, so no basis has to be guessed in advance. Dividing by $2 \sinh(\theta T)$ gives the hyperbolic interpolation; setting $X_0 = X$, $X_T = 0$ gives equation (4).

Trade List

Equation (5)

Since $n_j = x_{j-1} - x_j$, equation (4) gives

$$n_j = \frac{X}{\sinh(\theta T)} [\sinh(\theta(T-t_{j-1})) - \sinh(\theta(T-t_j))].$$

The identity $\sinh u - \sinh v = 2 \cosh((u+v)/2) \sinh((u-v)/2)$, with $u-v = \theta\tau$ and $(u+v)/2 = \theta(T-t_{j-\frac{1}{2}})$, gives equation (5).

Notation

symbol	meaning	units
X	total shares to be liquidated (or accumulated)	shares
X_0, X_T	holdings at the start and end of a schedule; $X_0 = X$ and $X_T = 0$ for a full liquidation	shares
T	trading horizon	day
$\tau = T/N$	length of one discretization step, with N the number of steps	day
$x_k, x(t)$	shares still held after step k , or at time t	shares
$S_k, S(t)$	security price after step k , or at time t , with S_0 the initial price	\$/share
α	drift rate of the price process; assumed zero throughout	\$/share/day
ξ_k	i.i.d. shock in step k , zero mean and unit variance	—
$n_k = x_{k-1} - x_k$	shares traded in step k	shares
n_k/τ	trading rate during step k : shares traded per unit time, not to be confused with n_k itself	shares/day
$g(v)$	function of trading rate v entering the price process; its effect carries forward into S_k	\$/share
$\tilde{S}_k$	actual price per share received on sale k	\$/share

symbol	meaning	units
$h(v)$	temporary-impact function: the drop in average price per share from trading at rate v during one interval; does not carry forward into S_k	\$/share
$E(x)$	expected cost of trading schedule x	\$
$V(x)$	variance of the cost of trading schedule x	$\2
$U(x) = E(x) + \lambda V(x)$	utility function minimized over the schedule; convex, so its unique minimum is found by setting $\partial U / \partial x_j = 0$	\$
γ	permanent-impact coefficient: the persistent price shift per share traded	\$/share ²
η	temporary-impact coefficient: the price concession per unit trading rate	\$/share ²
ε	fixed cost per trade in $h(v)$, e.g. half the bid-ask spread plus fees	\$/share
$\bar{\eta} = \eta - \frac{1}{2}\gamma\tau$	temporary-impact coefficient net of the permanent-impact schedule-dependent term; $\bar{\eta} \rightarrow \eta$ as $\tau \rightarrow 0$	\$/share ²
σ	volatility of the traded asset	\$/share/ $\sqrt{\text{day}}$
λ	risk aversion: the trader's chosen weight on cost variance	1/\$
$\tilde{\kappa} = \sqrt{\lambda\sigma^2/\bar{\eta}}$	coefficient of the discrete optimality recurrence	1/day
θ	decay rate of the optimal schedule, solving $2\tau^{-2}(\cosh(\theta\tau) - 1) = \tilde{\kappa}^2$; $\theta \rightarrow \kappa$ as $\tau \rightarrow 0$	1/day
$\kappa = \sqrt{\lambda\sigma^2/\eta}$	continuous-time limit of both θ and $\tilde{\kappa}$	1/day

γ and η each carry an extra factor of 1/ share beyond what “a price shift per share traded” suggests, because **price** is itself quoted in \$/share: a shift in that per-share price, per share traded, is (\$/share)/share = \$/share².